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José Manuel Viejo
Mariano Sanjuán *Editors*

Life and Mind

New Directions in the Philosophy of Biology and Cognitive Sciences

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Editors

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Incommensurability in Evolutionary Biology: The Extended Evolutionary Synthesis Controversy



Juan Gefaell and Cristian Saborido

Abstract Evolutionary biologists today debate whether it is convenient to revise the standard theory of evolution, or if an Extended Evolutionary Synthesis is necessary. However, the conceptual relationship between the standard theory of evolution (also known as the Modern Synthesis) and a putative Extended Evolutionary Synthesis is not clear. One concept in philosophy of science that has traditionally been put in place to make sense of the conceptual relationship between competing theories or frameworks is that of Kuhnian incommensurability. In a book chapter, Pigliucci argued that the Modern Synthesis and the Extended Evolutionary Synthesis are not incommensurable frameworks and that their relationship is best understood as a business-as-usual extension of our current knowledge about evolution. However, while valuable, we believe that Pigliucci's analysis is limited in several respects. After pointing out what these limitations are, in this chapter we try to provide an alternative analysis of incommensurability between the Modern Synthesis and the Extended Evolutionary Synthesis. We argue that there are compelling reasons to think that both frameworks are incommensurable, thereby leaving the door open for future philosophical explorations.

Keywords Modern Synthesis · Extended Evolutionary Synthesis · Thomas S. Kuhn · Incommensurability · Evolutionary biology

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1 Introduction

Current evolutionary biology is immersed in a far-reaching debate about the desirability of revising the standard theory of evolution, also known as the Modern Synthesis (MS) (e.g., Laland et al. 2014; Futuyma 2017; Müller 2017). Some evolutionary biologists claim that many empirical and theoretical advances made during the last decades in diverse branches of biology call for a supposedly 'extended' version of evolutionary theory, which they have accordingly called the Extended Evolutionary Synthesis (EES). Other evolutionary biologists are skeptical, though, of the EES, instead arguing that the MS can perfectly account for these empirical and theoretical advances without revising any of its fundamental tenets.

What is the exact relationship between the MS and the EES? Is the EES just an extension of the MS? Or, conversely, does it represent a more radical break with standard evolutionary theory? These questions are relevant not only to evolutionary biologists themselves but also to philosophers of science interested in how scientific theories change over time (Andersen and Hepburn 2013, online). Scientific change has been a classic topic of study for philosophers of science not only for its historical interest but also because it can give us clues about general questions related to the rationality of science.

One way to interpret scientific change is through the concept of incommensurability. Originally coined by Thomas S. Kuhn and Paul Feyerabend in 1962 (Kuhn 1970[1962], Feyerabend 1962), incommensurability can be defined as the lack of neutral standards from which to compare and assess the merits of competing theoretical frameworks. Incommensurability can be thought of as a proxy for a radical break; if, after analysing two competing theoretical frameworks, enough instances of incommensurability are found, then we can conclude that the transition between these frameworks is not continuous but abrupt. If, conversely, no real signs of incommensurability are detected, then we can argue that the analysed theoretical frameworks are not that different and that there is a significant degree of continuity between them.

In a book chapter, philosopher and evolutionary biologist Massimo Pigliucci has made use of the concept of incommensurability as seen by Kuhn to analyse the purported transition between the MS and the EES (Pigliucci 2017). After examining their conceptual structure, he concluded that the MS and the EES are not incommensurable and that the transition between them is best viewed as a business-as-usual extension rather than a radical break. However, his attempt, while valuable, is limited in some respects. Incommensurability as described in Kuhn's work is a complex concept that, in some cases, can lead to misinterpretations. We suggest that Pigliucci's use of incommensurability includes some of these misinterpretations, thus limiting the validity of his analysis of the MS versus EES controversy.

In this chapter, we try to show why Pigliucci's account is problematic and present an alternative analysis of incommensurability between the MS and the EES. Unlike Pigliucci, we believe that there are substantial reasons to think that the MS and the EES are incommensurable and that the EES does not probably constitute an ordinary

extension of the standard theory of evolution but a significantly different theoretical framework.

But before going any further, we must ask, however, the following question: Does Kuhn still have something meaningful to tell us about how scientific disciplines change? Many scholars would probably doubt if there is any point in employing a Kuhnian concept to analyse a scientific dispute. Today, Kuhn's ideas are often dismissed without due consideration by many philosophers of biology. This is especially the case in the philosophy of evolutionary biology, where Mayr's critique of Kuhn has convinced most philosophers that this author has nothing interesting to offer to understand the historical dynamics of evolutionary biology (Mayr 1994, 2004). However, in recent decades, the Kuhnian concept of incommensurability has experienced a resurge in popularity; many philosophers and historians of science have successfully used this concept to explain certain aspects of scientific disputes throughout the history of science (Wray 2005, 2011; Chang 2013; Politi 2018). In addition, Tanghe and collaborators have recently conducted a Kuhnian analysis of the history of evolutionary biology, concluding that 'interpreting [the] history [of evolutionary biology] through a Kuhnian prism may be somewhat unorthodox but it most definitely makes sense' (Tanghe et al. 2021: 25). Following this recent trend, in this chapter, we would like to show how a distinctively Kuhnian concept (incommensurability), when properly considered, may shed light on how debates in contemporary evolutionary biology unfold.

2 Pigliucci, Kuhn, and Incommensurability

Pigliucci has been one of the first scientists to call for an EES (Pigliucci 2007, 2009; Pigliucci and Müller 2010). However, since his first publications on the subject, he has made clear that this new and purportedly extended theory of evolution does not constitute a radical break with the past (Pigliucci 2007, 2009, 2012, 2017). To argue for this conclusion, in some of these publications this author has made use of Kuhnian concepts. More specifically, in a 2017 book chapter, Pigliucci (2017) used the notion of incommensurability as depicted in *The Structure of Scientific Revolutions* (*SSR*; Kuhn 1970) to argue that the MS and the EES are not incommensurable theoretical frameworks. But what exactly does that mean? To understand Pigliucci's claim, let us first take a brief look at incommensurability in *SSR*.

2.1 Incommensurability in SSR

Kuhn introduced incommensurability in *SSR* to account for the relationship between two competing paradigms that are involved in a scientific revolution (Kuhn 1970; see also Hoyningen-Huene 1990; Sankey 1993; Hoyningen-Huene and Sankey 2001). Kuhn understood paradigms—or *disciplinary matrices*, as he would later

call them—as all-encompassing guides for conducting science that tell scientists how to approach new scientific problems ('exemplars'), which theoretical principles should be employed to explain phenomena ('symbolic generalisations'), how to conceptualise the world ('models'), and which epistemic values a theory should satisfy ('values') (Kuhn 1970). According to Kuhn, these competing paradigms are so different from each other in terms of their exemplars, symbolic generalisations, models, and values that a neutral comparison between them is not possible. The assessment of their scientific merits is always based on the assumptions of one of the two competing paradigms. Kuhn referred to this impossibility of a neutral paradigm comparison as incommensurability.[1]

Many Kuhn scholars have argued that the *SSR* notion of incommensurability covers three different but interrelated domains of scientific practice: methodology, observations, and semantics (see, for instance, Hoyningen-Huene 1993). *Methodological* incommensurability amounts to disagreements over the kinds of problems that each paradigm considers relevant. For Kuhn, different paradigms have significantly different research agendas, up to the point of considering a rival's research agenda as irrelevant, obsolete or, in the most extreme cases, even unscientific (Kuhn 1970). In addition, methodological incommensurability also refers to disagreements over which theoretical or epistemic values should be prioritised when evaluating the scientific quality of a theory or hypothesis. Finally, methodological incommensurability also includes those differences that scientists of competing paradigms have about the ultimate aims of their scientific discipline: Should it aspire to a grand unified theory, or, on the contrary, should it focus on building partial models of phenomena?

Observational incommensurability is the process by which 'proponents of competing paradigms practice their trades in different worlds' (Kuhn 1970: 150). This vague definition of observational incommensurability has led to a great deal of speculation about the correct way of interpreting it. According to the most common interpretation, observational incommensurability would refer to a perceptual process by which observable entities and processes are 'seen' differently by proponents of each paradigm due to the theory-ladenness of observation.[2]

Finally, *semantic* incommensurability entails that scientists from different paradigms employ the same terms in significantly different ways, with different meanings and theoretical implications. For example, according to Kuhn (1970), the meaning and theoretical role of the concept of 'mass' in Newtonian and Einsteinian physics are radically different; when discussing the merits of their paradigms, Newtonian and Einsteinian physicists do not use the term 'mass' in the same way,

[1] In this chapter, we focus only on incommensurability as it is portrayed in *SSR*, not only because this is the notion of incommensurability that Pigliucci employs, but also because we believe it is much better suited to analyse scientific disputes than its taxonomic, post-*SSR* counterpart (for a comparison of both versions of incommensurability as applied to the MS versus EES controversy see Gefaell and Saborido (2022)).

[2] However, as we will see in Sect. 3.3, this is probably not the best way to interpret observational incommensurability.

although they use the same word. Semantic incommensurability causes communication problems between scientists of different paradigms.

In sum, according to Kuhn, incommensurability refers to the supposed absence of neutral grounds to compare competing paradigms at the level of their methodological, observational, and semantic domains.

2.2 What Does Pigliucci Say and Why Is It Problematic?

To establish whether the MS and the EES are incommensurable, in his book chapter, Pigliucci discusses the three aforementioned spheres of incommensurability (methods, observations, and semantics) as applied to this particular controversy, concluding that:

1. There is no methodological incommensurability between the MS and the EES because 'the EES takes on board the same puzzles [i.e., the same research agenda], and the same set of approaches, of the MS, but also adds new puzzles (. . .)' (Pigliucci 2017: 99).
2. There is no observational incommensurability because 'the very same facts that have been catalogued and explained by the MS enter into the empirical corpus of the EES' (Pigliucci 2017: 99).
3. There is no semantic incommensurability because '[k]ey biological concepts (. . .) retain similar and perfectly commensurable meanings' in both theoretical frameworks (Pigliucci 2017: 99).

Based on these assertions, Pigliucci argues that contemporary evolutionary biology is not undergoing a paradigm-shifting revolution, but rather a 'continuous expansion of both empirical knowledge and conceptual understanding' (Pigliucci 2017: 100), similar to those that, according to his view, would have taken place throughout the history of evolutionary biology since the Darwinian Revolution.

Pigliucci's analysis constitutes a valuable first assessment of the possible incommensurability between MS and EES. However, we believe that it is based on at least four dubious – or even downright incorrect – assumptions about incommensurability and the nature of the MS versus the EES controversy. More specifically:

(a) *Pigliucci seems to assume that incommensurability is a holistic phenomenon.*
(b) *Pigliucci's exposition of the debate between the MS and the EES overlooks the most contentious issues of the controversy.*
(c) *Pigliucci does not interpret observational incommensurability in the most meaningful way.*
(d) *Pigliucci assumes that incommensurability always implies a revolutionary process of paradigm shift.*

In the next section, we will address these four problems, showing how they compromise Pigliucci's main conclusion, namely that the MS and the EES are commensurable theoretical frameworks. In Sect. 4, we offer an alternative analysis

of incommensurability between the MS and the EES that seeks to overcome these problems in Pigliucci's analysis.

3 A Closer Look at the Problems of Pigliucci's Analysis

3.1 Incommensurability Is Not a Holistic Phenomenon

The first problematic assumption in Pigliucci's analysis is that he seems to assume that incommensurability is a global phenomenon, that is, a phenomenon that affects *all* methods, observations, and/or concepts of competing theoretical frameworks. Because Pigliucci finds at least some common methods, observations, and concepts between the MS and the EES, he concludes that both frameworks are not incommensurable. It is true that, in his chapter, Pigliucci asserts that if incommensurability is found in just one domain (either methodology, observations, or semantics), then 'a good argument can be made that a paradigm shift [i.e., incommensurability] is occurring' (Pigliucci 2017: 99). However, he seems to assume that *all* items in either of these classes have to be incommensurable in order for incommensurability to occur.[3]

We believe that this is a dubious interpretation of incommensurability. In our view, the fact that incommensurability can act as a proxy for radical departure does not imply that two incommensurable theoretical frameworks must be different in all of their methods, observations, and/or concepts, or that no points of continuity can be found between them. Even fully incommensurable paradigms share some items. This is so because, as Kuhn pointed out, incommensurability is not a holistic but a local phenomenon (Kuhn 2000a), meaning that it is generally restricted to a limited – yet especially important—set of items in each domain. Pigliucci seems to miss this point in his discussion of incommensurability, tacitly assuming that for incommensurability to occur there cannot be any overlap between the MS and the EES. A more thoughtfully Kuhnian interpretation must consider the local nature of incommensurability.

3.2 Analyses of Incommensurability Should Focus on the Most Controversial Aspects of the Dispute

The fact that incommensurability is a local phenomenon restricted to a limited – yet especially important – subset of methods, observations, and concepts force us to carefully reflect on which of these methods, observations, and concepts we should choose to perform our incommensurability analyses. Otherwise, the risk is to

[3] We thank an anonymous reviewer for pointing this to us.

conclude that two potentially incommensurable paradigms are commensurable just because we have focused on those methods, observations, or terms that both paradigms share. We believe that this is what happens in Pigliucci's analysis: When arguing against incommensurability, he focuses on the relatively uncontroversial parts of the MS versus EES debate, giving the impression that there is much more continuity between these two frameworks than there is. For example, two terms that according to Pigliucci show that the MS and the EES are semantically commensurable are 'species' and 'phenotype'. However, the fact that these terms are not under dispute does not mean that there is also agreement on the meaning of other crucial concepts.

Because of this, we believe that analyses of incommensurability should focus only on the most controversial items in a given scientific debate. Only by analysing the most controversial methods, observations, and concepts would we be able to determine whether there is incommensurability between the two theoretical frameworks involved. To detect what these items are, it is important to look not only at the conceptual structure of the theoretical frameworks involved as they are depicted in scientific publications but also at the way scientists from different paradigms debate in scientific meetings and confrontational papers. This is important because some methods, observations, or concepts that from a strictly theoretical point of view may not seem very problematic may turn out to be so when we observe how scientists debate about them. In fact, it could be argued that this is yet another problem with Pigliucci's analysis: he does not refer to the growing number of scientific papers in which proponents of the MS and the EES debate different points of their theories (Dickins and Rahman 2012; Mesoudi et al. 2013; Scott-Phillips et al. 2014; Laland et al. 2014; Gupta et al. 2017a, b; Feldman et al. 2017; Welch 2017; Laland 2017; Müller 2017; Futuyma 2017; Tanghe et al. 2018; Svensson 2023). These may give a better impression as to whether the MS and the EES are really incommensurable than a purely theoretical analysis. Although Kuhn did not specify where exactly to look for conducting incommensurability analyses, we believe that our proposal is fully in line with a Kuhnian view of science, since it places its emphasis on scientific practice rather than only on its purely formal elements (Rouse 2003).

3.3 Observational Incommensurability Is Really About Ontological Assumptions

The third problematic assumption in Pigliucci's analysis is related to his interpretation of observational incommensurability. Pigliucci seems to adhere to the most common interpretation of observational incommensurability, arguing that it means that 'what is considered a "fact" within one theoretical context may not be such in a different theoretical context' (Pigliucci 2017: 99). As we suggested in Sect. 2.1, there is no straightforward way to interpret observational incommensurability, given Kuhn's ambiguities on that matter, so Pigliucci cannot be blamed for interpreting

observational incommensurability in such a way. However, we believe that there is a more suggestive and Kuhnian way of interpreting observational incommensurability than that of Pigliucci.

According to this interpretation, observational incommensurability would amount to irreconcilable differences in very basic ontological assumptions; assumptions about which entities and processes populate the world. As Kuhn pointed out, each paradigm or theoretical framework includes a series of fundamental metaphysical beliefs that guide the interpretation of directly observable phenomena (i.e., 'models'). The ontological assumptions of competing paradigms depart significantly from each other to the point of being incompatible. For this reason, controversies regarding observational incommensurability do not revolve around 'what is considered a "fact"', as Pigliucci and many others suggest, but rather around how—that is, under which ontological assumptions—to interpret a *shared* set of facts. Due to its emphasis on ontological assumptions rather than empirical facts, we have elsewhere called this interpretation of observational incommensurability 'ontological incommensurability' (Gefaell and Saborido 2022).

This ontology-centred way of understanding observational incommensurability is supported by some examples discussed by Kuhn in *SSR*, such as that of the conceptions of space in Newtonian and relativistic physics. According to Kuhn, Newtonian scientists are 'embedded in a flat [space]', while Einsteinians inhabit '(. . .) a curved matrix of space' (Kuhn 1970: 150). The concept of space is not an observable phenomenon, but a very broad ontological category, and beliefs in a 'flat' or 'curved matrix' of space are not empirical claims, but metaphysical beliefs or assumptions that guide the interpretation of data collected by scientists. This example suggests that Kuhn understood observational incommensurability in a much more ontology-oriented way than traditional accounts of the concept assume. Recent attempts to develop observational incommensurability under a connectionist framework, such as that of Alexander Bird (2005, 2008), are also congruent with this interpretation of observational incommensurability as referring to the ontological assumptions of paradigms. We suggest that this way of interpreting observational incommensurability is not only closer to Kuhn's original position, but also may yield more interesting results when applied to scientific controversies.

3.4 Incommensurability Does Not Imply Paradigm-Shifting Revolutions

The fourth problematic assumption in Pigliucci's analysis is that he automatically links incommensurability with paradigm-shifting revolutions. However, a closer look at Kuhn's writings reveals that incommensurability does not necessarily imply a paradigm-shifting revolution, but it can also lead to other kinds of scientific change, particularly the formation of new scientific specialties (Kuhn 2000b, c; Politi

2019). This means that it is possible to argue in favour of incommensurability without committing to the inevitability of a paradigm shift.

Instead of a paradigm-shifting revolution, in the scientific specialisation scenario incommensurability leads to the splitting of a scientific community into two coexisting, yet completely autonomous, scientific communities. According to the most common interpretation, this process of scientific specialisation is normally triggered by anomalies (see Politi 2019). As Kuhn explains in *SSR*, at the time of their inception, all paradigms have anomalies or unsolved problems. Most of the time, these anomalies are overlooked by the majority of ordinary scientists, who instead focus on easier problems that do not call into question the established framework (what Kuhn called 'puzzles'), leaving anomalies for a time when the paradigm is ripe enough to resolve them. However, a minority of scientists still spend a great deal of time thinking and investigating these anomalies. This may lead to the development of new tools, concepts, or hypotheses, some of which may divert from the 'orthodox' point of view. If some of these tools, concepts, or hypotheses shed light onto some anomalies, then it may be the starting point of a specialisation process: they may further the divergence with established methods and concepts and may speed up the discovery of new phenomena that cannot be accounted for under the orthodox framework. On these occasions, the minority of scientists who initiated the above-mentioned changes may eventually end up becoming a separate scientific community, with its own tools and research agenda. When, after some time, this newly formed scientific community seeks to establish contact with its 'mother' community, differences in methods, assumptions, and concepts would prevent full communication; in other words, incommensurability would occur. In this sense, incommensurability would be the 'isolation mechanism' by which both communities lose contact with each other.[4]

Given that incommensurability can be associated with different kinds of scientific change, if one obtains evidence in favour of incommensurability, it becomes necessary to determine independently what mode of scientific change (paradigm shift, scientific specialisation, or other) accompanies incommensurability in that particular case. Pigliucci is wrong in automatically assuming that incommensurability necessarily implies a paradigm-shifting revolution.

[4]The reference to biological vocabulary, such as 'isolation mechanism', is no coincidence, since Kuhn employed the 'speciation' metaphor to understand scientific specialisation. As in allopatric speciation, where one species eventually becomes two separate ones after it is split up in two populations by a geographical barrier, making selective pressures to act differently in each population, scientific specialisation would also involve an isolation mechanism (incommensurability) and a divergence in selective pressures (different problems tackled by each group of scientists).

4 An Alternative Analysis of Incommensurability Between the MS and the EES

Due to the aforementioned problems in his account, Pigliucci asserts that there is no incommensurability between the MS and the EES at the methodological, observational, and semantic levels. Do we get the same conclusion when we try to avoid these problems? We argue that this is not the case: we believe that when these problems are addressed, evidence suggests that the MS and the EES are incommensurable theoretical frameworks. To show why, let us follow the same approach as Pigliucci, considering methods, observations, and concepts separately.[5]

4.1 Methodological Incommensurability Between the MS and the EES

We have already seen that Pigliucci defends that there is no methodological incommensurability between the MS and the EES. We disagree, as we detect at least four instances of methodological incommensurability between these two theoretical frameworks.

The first one has to do with their research agenda, or, in other words, the kinds of problems that each theoretical framework considers of high priority and most scientific interest. The MS research agenda is markedly influenced by population genetics. MS evolutionary biologists are concerned with, for instance, problems related to effective census (*Ne*) estimation, the modeling of genetic drift on allele frequencies, or the effects of consanguinity on the average fitness of populations, to name a few examples. In this sense, we believe most adherents to the MS still agree with Dobzhansky's claim that '(. . .) the mechanisms of evolution constitute problems of population genetics' (Dobzhansky 1937: 11–12).[6]

On the contrary, supporters of the EES do not grant population genetics such importance in their research agenda. Instead, they tend to focus on developmental processes and their role in evolution, as well as what can be called 'agentic' problems, that is, those that have to do with intentional or agential behaviours performed by organisms and their impact on evolution (such as, for example,

[5] We will also follow Pigliucci (2017) in understanding the EES as the framework exposed in Laland et al. (2015), as we agree with him that this paper 'is both more focused and more systematic [in its depiction of the EES] than previous attempts' (Pigliucci 2017: 96). As for the MS, we understand it as the theoretical framework depicted in mainstream evolutionary biology textbooks, such as Ridley (2004), Freeman and Herron (2004), or Futuyma (2009). This move is not a coincidence, since Kuhn stressed the role of textbooks in establishing dominant paradigms (Kuhn 1963).

[6] Other more recent vindications of the importance of population genetics for the MS can be seen in Ridley (2004: 93), Freeman and Herron (2004: 141), Lynch (2007), or Futuyma (2009: 220).

many actions and processes that fall under the label of 'niche construction'; see Odling-Smee et al. 2003; Laland et al. 2015). The different research agendas of each framework also influence the kinds of organisms that are used to conduct research in the MS and the EES. In this sense, the EES is much more open to novel model organisms that are better suited to the study of developmental processes, such as *Onthophagus* beetles (Baedke et al. 2020), whereas advocates of the MS rely more on traditional model organisms, such as *Drosophila* flies or guppy fishes. Of course, these differences in their research agendas and model organisms are far from absolute. We are not arguing that supporters of the EES completely disregard population genetics or that the MS gets no interest at all in developmental processes, but that there is a major shift of emphasis in their research priorities. We believe that this shift is of much more relevance than Pigliucci thinks and that it is responsible for many conflicts between proponents of the MS and the EES.

The second instance of methodological incommensurability has to do with what we have called *explanatory preferences* (Gefaell and Saborido 2022). Explanatory preferences can be defined as the kinds of entities and processes that a given theoretical framework prioritises as *explanantia* in scientific explanations. In the life and mind sciences, there are two widely known explanatory preferences: explanatory externalism and explanatory internalism (Godfrey-Smith 1996). Explanatory externalism refers to 'explanations of properties of organic systems in terms of properties of their environments', while explanatory internalism refers to '[e]xplanations of one set of organic properties [in this case, the capacity for evolution] in terms of other internal or intrinsic properties of the organic system' (Godfrey-Smith 1996: 30).

Following these definitions, we believe that the MS leans toward explanatory externalism, while the EES toward explanatory internalism (e.g., Fábregas-Tejeda and Vergara-Silva 2018). This difference in explanatory styles can be seen, for instance, in the way each theoretical framework explains adaptations, that is, the fit between organisms and their environments: While the MS usually conceives these as the direct result of selective pressures imposed on organisms by – external – ecological factors, the EES emphasises the role of developmental plasticity and niche construction along natural selection in achieving such a fit (Scott-Phillips et al. 2014; Laland et al. 2015). This move from explanatory externalism to explanatory internalism is acknowledged by proponents of the EES themselves:

> '(. . .) [the EES] moves the focus of evolutionary explanation from the external and contingent to the internal and inherent. It posits that the causal basis for phenotypic form resides not in population dynamics or, for that matter, in molecular evolution, but instead in the inherent properties of evolving developmental systems' (Müller 2007: 947–948).

The third instance of methodological incommensurability between the MS and the EES involves epistemic values. Kuhn argued that although all natural scientists, *qua* scientists, share a series of epistemic values that are constitutive of science itself, their ranking, or the relative importance granted to each of them, varies between theoretical frameworks (Kuhn 1977). This observation also holds true for the MS and the EES. For instance, the relative importance given to the epistemic value of

simplicity is different in the MS and the EES: supporters of the MS tend to give this value more importance than advocates of EES. An example of this can be seen in Douglas Futuyma's claim that 'simpler explanations are generally preferred over more complex (and vague) hypotheses, unless these are supported by evidence' (Futuyma 2017: 8), referring to EES-styled explanations. Compare this statement with advocates of the EES claiming that explanations of the MS are often 'too simple' or 'impoverished', in a clear derogation of the value of simplicity (Scott-Phillips et al. 2014: 1233–1240). Rather than promoting simplicity, advocates of the EES grant less idealised explanations a higher epistemic value—even if these explanations are much more difficult to model because they involve many more causal factors than those of the MS.[7]

Finally, the fourth instance of methodological incommensurability has to do with the different views each theoretical framework has about the aims and general outlook of evolutionary biology as a science. In this regard, advocates of the EES have a much more favourable opinion toward scientific pluralism than their MS counterparts. Several quotes illustrate this; for example: 'We believe that a plurality of perspectives in science encourages the development of alternative hypotheses, and stimulates empirical work' (Laland et al. 2014: 164); 'We believe that a plurality of perspectives in science is healthy (. . .)' (Laland et al. 2015: 10; see also Laland 2018; Uller et al. 2019). On the contrary, adherents to the MS, with its insistence that a new evolutionary theory is not necessary (even when supporters of the EES explicitly say that they do not want to supersede the MS, but only to turn evolutionary biology into a plural discipline), seem to be much more inclined toward scientific monism or the unity of science. Theoretical unification was a major epistemic goal during the inception of the MS (Smocovitis 1992), and it still probably is for many orthodox evolutionary biologists (but see Svensson 2023).

4.2 Observational (That Is, Ontological) Incommensurability Between the MS and the EES

Although the observational basis of the MS and the EES is mostly shared, the ontological assumptions underlying their interpretation of phenomena are significantly different and in many cases difficult to reconcile. We detect at least two instances of ontological incommensurability: one related to the nature of organisms and the other to the nature of causality. Regarding the nature of organisms, the MS and the EES assume two significantly different metaphysical beliefs: On the one hand, the ontology of the MS is gene-centred, reflecting the population genetic basis

[7] This can be seen, for instance, in Uller et al. (2019). These authors contrast evolutionary explanations based on natural selection acting on genes with evolutionary explanations based on developmental plasticity, physiology, and behaviour. The latter explanations include many more causal factors, and assume more complex causal chains, than the former.

of the MS (Provine 1971). The MS conceives organisms as gene-carriers, or, as Richard Dawkins has put it, 'vehicles' for genes (Dawkins 1976; see also Hull 1980). Genes are the entities that ultimately evolve, not organisms.[8] On the other hand, for the EES, organisms occupy the centre stage. Organisms are not reducible to genes and instead act as integrated wholes. They share a set of special properties that, although not incompatible with physical laws, transcend them. They have the capacity for agency and autonomy and, for at least some EES-friendly authors, cannot be studied under a purely mechanical framework (e.g., Walsh 2015). The organicism espoused by the EES has several theoretical and methodological consequences. Particularly, it can be argued that the fact that adherents of the EES confer the study of developmental biology and agentic behaviour such importance stems from their metaphysical assumption that organisms are integrated wholes with special properties which are not reducible to mechanical ones.

The second instance of ontological incommensurability between the MS and the EES is the nature of causality (e.g., Laland et al. 2011, 2013; Martínez and Esposito 2014; Fábregas-Tejeda and Vergara-Silva 2018; Uller and Helanterä 2019). A core ontological assumption of the MS is the so-called proximate versus ultimate dichotomy of causation, originally put forward by one of the MS architects, Ernst Mayr, in the 1960s (Mayr 1961). This distinction implies that the proximate causes of biological traits (their developmental and physiological basis) have nothing to do with their evolutionary causes. Conversely, followers of the EES explicitly reject the proximate versus ultimate dichotomy. In fact, it can be argued that their entire research project is based on the premise that this dichotomy does not hold true and that what under such a framework are considered proximate mechanisms can indeed have a lasting influence on evolution (Laland et al. 2011, 2013, 2015).

Many misunderstandings that surround the debate around the convenience of the EES involve the status of the proximate versus ultimate dichotomy. When proponents of the EES vindicate the importance of development and behaviour for evolution, supporters of the MS reply that these are not relevant processes for a theory of evolution because they constitute proximate causes (e.g., Dickins and Rahman 2012; Futuyma 2017). The EES camp rejects this charge because they deny the validity of such a dichotomy (e.g., Laland et al. 2013, 2015). This generates fierce disagreements and communication failures between both camps.

In addition to the dispute over the status of the proximate versus ultimate dichotomy, other ontological assumptions regarding the nature of causation also generate disagreement between the MS and the EES. These include multilevel causation (Noble 2012; Martínez and Esposito 2014), and the so-called 'fragmentation of evolution', the notion tacitly assumed by supporters of the MS according to which the processes involved in inheritance, variation, and differential fitness are

[8] It could be replied that the gene-centred ontology was disputed by some of the early proponents of the MS, notably Ernst Mayr (1963). However, even authors such as Mayr can be seen as committed to a gene-centred view of the organism. This is so because although Mayr has been known for his criticism of 'bean-bag genetics', he has also argued that organisms are the result of the unfolding of a genetic program (Mayr 2004).

causally independent. Some proponents of the EES reject this fragmentation and claim that all of these processes are causally intertwined (Walsh 2015, 2016; Uller and Helanterä 2019).

4.3 *Semantic Incommensurability Between the MS and the EES*

Although we agree with Pigliucci that the meaning of many terms is shared by the MS and the EES, we believe that other concepts are more problematic and cast doubt on Pigliucci's conclusion. For example, a potential example of semantic incommensurability between these theoretical frameworks is the very concept of 'evolution'. Under the MS, evolution is understood as changes in gene or allele frequencies (e.g., Dobzhansky 1937; Freeman and Herron 2004; Futuyma 2009). The EES, for its part, usually employs (supposedly) much broader definitions of evolution, such as, 'transgenerational change in the distribution of heritable traits of a population' (e.g., Laland et al. 2015: 2). These differences in the way each theoretical framework defines a concept as important as 'evolution' sometimes gives rise to communication failures. For example, for many supporters of the MS, niche construction cannot count as an evolutionary mechanism (as advocates of the EES believe) because it does not alter gene frequencies (Scott-Phillips et al. 2014). On the contrary, given that the EES defines evolution in an allegedly much broader way, including changes in ecological inheritance brought up by niche construction, its advocates have no problem in considering this process as an evolutionary mechanism.[9]

Some may object that what we have just argued is perfectly compatible with Pigliucci's account, as the examples discussed above suggest that evolution under the EES is understood in a *much broader way than* – yet *not incommensurable with* – the corresponding concept in the MS. To this objection, our reply is that, despite appearances, the meaning of evolution in the EES is not a mere extension of that concept in the MS. The reason why this is so is that each conception of evolution carries with it significantly different metaphysical assumptions (particularly, those discussed in the previous section). More specifically, 'evolution' in the MS assumes the causal independence of inheritance, variation, and differential fitness (the 'fragmentation of evolution'; Walsh 2015, 2016; Uller and Helanterä 2019), while the EES stresses their interactions. For this reason, a theoretical integration between the notions of 'evolution' of the MS and the EES is difficult and would require a great deal of conceptual work.

In sum, contrary to Pigliucci, we believe that there are several cases of methodological, observational (i.e., ontological), and semantic incommensurability between

[9] Similar deep disagreements can be found with respect to other important evolutionary concepts, such as, for instance, 'inheritance' (Danchin et al. 2011; Danchin and Pocheville 2014; Uller and Helanterä 2017; see Gefaell and Saborido 2022).

Table 1 A summary of cases of incommensurability between the MS and the EES

Incommensurability between the MS and the EES	
Methodological	Different research agenda
	Different explanatory preferences
	Different ranking of epistemic values
	Different views about science itself
Observational (i.e., ontological)	Different views on organisms
	Different views on causality
Semantic	Different understanding of evolution and other central concepts (particularly inheritance)

the MS and the EES. As for the methods, these frameworks differ in their research agenda, explanatory preferences, epistemic values, and view of science. With respect to ontology, the MS and the EES have significantly different views on the nature of organisms and causality. Finally, both frameworks espouse different notions of evolution (see Table 1).

5 What Type of Scientific Change Is Currently Going on?

So far, our analysis has left the question of which mode of scientific change accompanies incommensurability unanswered. In Sect. 3.4, we argued that from a Kuhnian point of view, incommensurability can prompt either a paradigm-shifting revolution or a process of scientific specialisation. So, which of the two is more likely to occur as a consequence of the controversy between the MS and the EES?

Although we believe that it is too early to rule out any possibility, there are at least two reasons not to commit to a paradigm-shifting scenario. The first reason has to do with the reluctance of most evolutionary biologists to join the EES. We do not have statistical data on the percentage of evolutionary biologists sympathetic to the EES, but our guess is that they constitute a small minority. Many evolutionary biologists working under the MS do not even know about the existence of an alternative theoretical framework, the EES being much better known and far more widely discussed in philosophical circles than in purely scientific ones. Unless the proponents of the EES present spectacular new achievements and run large outreach campaigns, it is not likely that this situation of relative ignorance by most practicing evolutionary biologists will change in the short term. It seems that advocates of the EES have given up the task of pushing for its own agenda. Although Kuhn (1970) argued that groups affected by a paradigm shift often number around one hundred or fewer members, there is still a long way to 'convert' most evolutionary biologists to the EES. For a paradigm-shifting revolution to occur, most scientists of the old paradigm must eventually abandon it and join the new paradigm. And given current circumstances, this is unlikely to happen in the near future.

The second reason we have for doubting a scientific revolution is the very intentions of the supporters of the EES themselves: most of them have explicitly said that they do not want to replace the MS, but only to turn evolutionary biology into a plural science.[10] As we have seen in Sect. 4.1, most supporters of the EES are committed scientific pluralists; they believe in the value of the co-existence of different perspectives, meaning that they only want to have their space within evolutionary biology alongside that of the MS to conduct their own research agenda, with their very own methods, concepts, and assumptions. This is even acknowledged by some critics of the EES (Svensson 2023). At least for now, we have no compelling reasons to think that these are not their real intentions. Therefore, if the supposedly 'revolutionary' scientists do not want to break with the MS, then the scenario of a paradigm shift weakens further.

But now we may ask: Does the above necessarily imply that scientific specialisation is going to take place? Although it might certainly seem more plausible at first sight and although it is compatible with the pluralist leanings of advocates of the EES, it is nonetheless too soon to conclude that the EES will turn into a separate specialty within the realm of evolutionary biology, perhaps engulfing evo-devo and other related subdisciplines. Besides, there is at least one potential caveat against this possibility, namely that the MS and the EES, far from being completely autonomous from each other in terms of their objects of study, in some respects compete for the explanation of the same evolutionary events. This can be seen in the case of niche construction, where the EES and the MS compete for the explanation of phenomena such as, for instance, the evolution of lactose tolerance in humans (Scott-Phillips et al. 2014). If two theoretical frameworks compete for the explanation of the same phenomena, then it is highly implausible that they will end up being completely autonomous scientific specialties. Points of friction and controversy will likely emerge.[11] This obliges us to remain agnostic about which kind of scientific change may follow from the incommensurability between the MS and the EES.

6 Conclusion

In this chapter, we have critically analysed Pigliucci's recent proposal to apply the Kuhnian notion of incommensurability to the MS versus EES controversy. We have argued that, while valuable, his analysis is based on four problematic assumptions

[10] Only a few advocates of the EES or EES-friendly authors adopt a more radical stance, arguing in favour of the substitution of the MS by an EES or equivalent (e.g., Noble 2015).

[11] A third plausible scenario that has been put forward recently is to conceive the EES as a Kuhnian reformulation of the MS (Tanghe et al. 2021: 15). This entails a sort of middle-ground between a paradigm-shifting revolution and a business-as-usual extension of the established paradigm. Viewing the EES as a Kuhnian reformulation of the MS is compatible with a local form of incommensurability like the one we have suggested in this chapter. However, what a Kuhnian reformulation entails and how to precisely detect it in particular contexts is something not yet clear.

that call into question his main conclusion, i.e., that there is no incommensurability between the MS and the EES. After showing what these problematic assumptions are, we have outlined our own analysis of incommensurability, concluding that there are several instances of incommensurability between the EES and the MS. However, we have also argued that our analysis of incommensurability does not imply that a paradigm shift is currently taking place in evolutionary biology.

If we are correct in our analysis, the existence of incommensurability between the MS and the EES leaves the door open for a number of additional philosophical problems that demand resolution. These include the nature and implications of deep disagreements in science, the rationality of theory evaluation, or the question of scientific realism (Psillos 2018). Elsewhere we have outlined potential avenues for the resolution of at least some of these problems (Gefaell and Saborido 2022), but a full exploration of them should be a matter of future investigation for philosophers and philosophically-inclined evolutionary biologists.

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References

Andersen H, Hepburn B (2013) Scientific change. In: The internet encyclopedia of philosophy. https://iep.utm.edu/scientific-change/. Accessed 30 May 2022

Baedke J, Fábregas-Tejeda A, Vergara-Silva F (2020) Does the extended evolutionary synthesis entail extended explanatory power? Biol Philos 35:20

Bird A (2005) Naturalizing Kuhn. Proc Aristot Soc 105:109–127

Bird A (2008) Incommensurability naturalized. In: Soler L, Sankey H, Hoyningen-Huene P (eds) Rethinking scientific change and theory comparison. Springer, Dordrecht, pp 21–39

Chang H (2013) Incommensurability: revisiting the chemical revolution. In: Kindi V, Arabatzis T (eds) Kuhn's structure of scientific revolutions revisited. Routledge, pp 153–178

Danchin E, Charmantier A, Champagne FA, Mesoudi A, Pujol B, Blanchet S (2011) Beyond DNA: integrating inclusive inheritance into an extended theory of evolution. Nat Rev Genet 12:475–486

Danchin E, Pocheville A (2014) Inheritance is where physiology meets evolution. J Physiol 592: 2307–2317

Dawkins R (1976) The selfish gene. Oxford University Press, New York

Dickins TE, Rahman Q (2012) The extended evolutionary synthesis and the role of soft inheritance in evolution. Proc R Soc B Biol Sci 279(1740):2913–2921

Dobzhansky T (1937) Genetics and the origin of species. Columbia University Press, New York

Fábregas-Tejeda A, Vergara-Silva F (2018) Hierarchy theory of evolution and the extended evolutionary synthesis: some epistemic bridges, some conceptual rifts. Evol Biol 45:127–139

Feldman MW, Odling-Smee J, Laland KN (2017) Why Gupta et al.'s critique of niche construction theory is off target. J Genet 96:505–508

Feyerabend PK (1962) Explanation, reduction and empiricism. In: Feigl H, Maxwell G (eds) Scientific explanation, space and time. Minnesota studies in the philosophy of science, vol 3. University of Minnesota Press, Minnesota, pp 28–97
Freeman S, Herron JC (2004) Evolutionary analysis. Pearson, New Jersey
Futuyma DJ (2009) Evolution. Sinauer Associates, Sunderland
Futuyma DJ (2017) Evolutionary biology today and the call for an extended synthesis. Interface Focus 7:20160145
Gefaell J, Saborido C (2022) Incommensurability and the extended evolutionary synthesis: taking Kuhn seriously. Eur J Philos Sci 12:24
Godfrey-Smith P (1996) Complexity and the function of mind in nature. Cambridge University Press, Cambridge
Gupta M, Prasad NG, Dey S, Joshi A, Vidya TNC (2017a) Niche construction in evolutionary theory: the construction of an academic niche? J Genet 96:491–504
Gupta M, Prasad NG, Dey S, Joshi A, Vidya TNC (2017b) Feldman et al. do protest too much, we think. J Genet 96:509–511
Hoyningen-Huene P (1990) Kuhn's conception of incommensurability. Stud Hist Philos Sci Part A 21:481–492
Hoyningen-Huene P (1993) Reconstructing scientific revolutions. Thomas Kuhn's philosophy of science. The University of Chicago Press, Chicago
Hoyningen-Huene P, Sankey H (eds) (2001) Incommensurability and related matters. Springer
Hull DL (1980) Individuality and selection. Annu Rev Ecol Syst 11:311–332
Kuhn TS (1963) The function of dogma in scientific research. In: Crombie A (ed) Scientific change. Heineman Educational Books, London, pp 347–369
Kuhn TS (1970) The structure of scientific revolutions. The University of Chicago Press, Chicago
Kuhn TS (1977) The essential tension: selected studies in scientific tradition and change. The University of Chicago Press, Chicago
Kuhn TS (2000a) Commensurability, comparability, communicability. In: Conant J, Haugeland J (eds) The road since *the structure*. The University of Chicago Press, Chicago, pp 33–57
Kuhn TS (2000b) The trouble with the historical philosophy of science. In: Conant J, Haugeland J (eds) The road since *the structure*. The University of Chicago Press, Chicago, pp 105–120
Kuhn TS (2000c) Afterwards. In: Conant J, Haugeland J (eds) The road since *the structure*. The University of Chicago Press, Chicago, pp 224–252
Laland KN (2017) Schism and synthesis at the Royal Society. Trends Ecol Evol 32:316–317
Laland KN (2018) Evolution unleashed. Aeon. https://aeon.co/essays/science-in-flux-is-a-revolution-brewing-in-evolutionary-theory. Accessed 26 Oct 2021
Laland KN, Odling-Smee J, Hoppitt W, Uller T (2013) More on how and why: cause and effect in biology revisited. Biol Philos 28:719–745
Laland KN, Sterelny K, Odling-Smee J, Hoppitt W, Uller T (2011) Cause and effect in biology revisited: is Mayr's proximate-ultimate dichotomy still useful? Science 334:1512–1516
Laland KN, Uller T, Feldman MW et al (2014) Does evolutionary theory need a rethink? Nature 514:161–164
Laland KN, Uller T, Feldman MW et al (2015) The extended evolutionary synthesis: its structure, assumptions and predictions. Proc R Soc B Biol Sci 282:20151019
Lynch M (2007) The frailty of adaptive hypotheses for the origins of organismal complexity. Proc Natl Acad Sci USA 104:8597–8604
Martínez M, Esposito M (2014) Multilevel causation and the extended synthesis. Biol Theory 9: 209–220
Mayr E (1963) Animal species and evolution. Harvard University Press, Harvard
Mayr E (1961) Cause and effect in biology. Science 134:1501–1506
Mayr E (1994) The advance of science and scientific revolutions. J Hist Behav Sci 30:328–334
Mayr E (2004) What makes biology unique? Cambridge University Press, Cambridge
Mesoudi A, Blanchet S, Charmantier A et al (2013) Is non-genetic inheritance just a proximate mechanism? A corroboration of the extended evolutionary synthesis. Biol Theory 7:189–195

Müller GB (2007) Evo-devo: extending the evolutionary synthesis. Nat Rev Genet 8:943–949
Müller GB (2017) Why an extended evolutionary synthesis is necessary. Interface Focus 7: 20170015
Noble D (2012) A theory of biological relativity: no privileged level of causation. Interface Focus 2: 55–64
Noble D (2015) Evolution beyond neo-Darwinism: a new conceptual framework. J Exp Biol 218:7–13
Odling-Smee J, Laland KN, Feldman MW (2003) Niche construction: the neglected process in evolution. Princeton University Press, Princeton
Pigliucci M (2007) Do we need an extended evolutionary synthesis? Evolution 61:2743–2749
Pigliucci M (2009) An extended synthesis for evolutionary biology. Ann N Y Acad Sci 1168:218–228
Pigliucci M (2012) Biology's last paradigm shift. Paradigmi 3:45–58
Pigliucci M (2017) Darwinism after the modern synthesis. In: Delisle RG (ed) The Darwinian tradition in context: research programs in evolutionary biology. Springer International Publishing, pp 89–103
Pigliucci M, Müller GB (eds) (2010) Evolution, the extended synthesis. The MIT Press, Cambridge
Politi V (2018) Scientific revolutions, specialization and the discovery of the structure of DNA: toward a new picture of the development of the sciences. Synthese 195:2267–2293
Politi V (2019) Specialisation and the incommensurability among scientific specialties. J Gen Philos Sci 50:129–144
Provine WB (1971) The origins of theoretical population genetics. The University of Chicago Press, Chicago
Psillos S (2018) Realism and theory change in science. The Stanford Encyclopedia of Philosophy. https://plato.stanford.edu/entries/realism-theory-change/. Accessed 31 May 2022
Ridley M (2004) Evolution. Wiley-Blackwell, Oxford
Rouse J (2003) Kuhn's philosophy of scientific practice. In: Nickles T (ed) Thomas Kuhn. Cambridge University Press, Cambridge, pp 101–121
Sankey H (1993) Kuhn's changing concept of incommensurability. Br J Philos Sci 44:759–774
Scott-Phillips TC, Laland KN, Shuker DM et al (2014) The niche construction perspective: a critical appraisal. Evolution 68:1231–1243
Smocovitis VB (1992) Unifying biology: the evolutionary synthesis and evolutionary biology. J Hist Biol 25:1–65
Svensson EI (2023) The structure of evolutionary theory: beyond neo-Darwinism, neo-Lamarckism and biased historical narratives about the modern synthesis. In: Dickins TE, Dickins BJA (eds) Evolutionary biology: contemporary and historical reflections upon core theory. Springer, pp 173–217
Tanghe KB, De Tiège A, Pauwels L, Blancke S, Braeckman J (2018) What's wrong with the modern evolutionary synthesis? A critical reply to Welch (2017). Biol Philos 33:23
Tanghe KB, Pauwels L, De Tiège A, Braeckman J (2021) Interpreting the history of evolutionary biology through a Kuhnian prism: sense or nonsense? Perspect Sci 29:1–35
Uller T, Feiner N, Radersma R et al (2019) Developmental plasticity and evolutionary explanations. Evol Dev 2:47–55
Uller T, Helanterä H (2017) Heredity and evolutionary theory. In: Walsh D, Huneman P (eds) Challenging the modern synthesis: adaptation, development, and inheritance. Oxford University Press, Oxford, pp 280–316
Uller T, Helanterä H (2019) Niche construction and conceptual change in evolutionary biology. Br J Philos Sci 70:351–375
Walsh DM (2015) Organisms, agency, and evolution. Cambridge University Press, Cambridge
Walsh DM (2016) Challenges to evolutionary theory. In: Humphreys P (ed) The Oxford handbook of philosophy of science. Oxford University Press, Oxford, pp 671–694
Welch JJ (2017) What's wrong with evolutionary biology? Biol Philos 32:263–279
Wray KB (2005) Rethinking scientific specialization. Soc Stud Sci 35:151–164
Wray KB (2011) Kuhn's evolutionary social epistemology. Cambridge University Press, Cambridge